

Mikhail Razakov

Compiler & Program Analysis Engineer

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LinkedIn

Compiler and program-analysis engineer with professional LLVM 22 experience at MCST and C# static analysis at ISP RAS / SharpChecker. Public projects demonstrate conservative interprocedural LLVM legality, Roslyn fixed-point data flow, and x86-64 code generation/register allocation.

EXPERIENCE

- 2026 — present** **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#.NET.
- July — August 2026** **MCST — Compiler Engineering Intern**
- Implemented LLVM 22 LICM with loop analysis and side-effect/speculative-safety checks.
 - Developed a conservative interprocedural Global-IV prototype with APInt affine evolution, transitive call effects, separate legality, and unsupported-case rejection.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv, verify, structural expectations, idempotence, and observable before/after execution comparison.

DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

x86-64 Codegen & Register Allocation Lab

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

SKILLS

Compiler / IR

C++23, LLVM 22/IR, APInt, interprocedural analysis, CFG/SSA, dominators, loops, LICM, legality

Program Analysis

C#.NET, Roslyn ControlFlowGraph, fixed-point data flow, conditional edges, loops/back edges, joins

Backend / Codegen

Rust, SysV x86-64, liveness/interference, register allocation, spills, phi lowering

Correctness / Tooling

LLVM Verifier, conservative rejection, regression checks, differential execution, Linux, CMake

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote